DISCRETE MATH · SET THEORY

Venn Diagram & Membership Table Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Venn Diagram & Membership Table Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. U = {1, …, 20}, A = evens, B = multiples of 3. Predict A ∩ B and (A ∪ B)′.
   A ∩ B = {6, 12, 18}; (A ∪ B)′ = {1, 5, 7, 11, 13, 17, 19}.
2. Is (A ∪ B)′ the same set as A′ ∪ B′? Predict, then check.
   No. (A ∪ B)′ = A′ ∩ B′ (De Morgan); A′ ∪ B′ = (A ∩ B)′ is larger.

2 Explore use the tool

1. Shade A ∩ B′ yourself, then type it as Expression 1. List its elements.
   {2, 4, 8, 10, 14, 16, 20}.
2. Make Expression 1 = (A ∪ B)′. Find an Expression 2 with ∩ and no ∪ that is equal. Which rows of the membership table did you check?
   A′ ∩ B′. All 4 rows agree (only the row "outside A and B" is ∈).
3. Write A ⊕ B without ⊕ and list its elements.
   (A ∪ B) – (A ∩ B) or (A – B) ∪ (B – A): {2, 3, 4, 8, 9, 10, 14, 15, 16, 20}.
4. Three sets (C = primes): shade the elements in exactly one of A, B, C. List them.
   {4, 5, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 19, 20}.

3 Record use the tool and fill in the table

Membership table: is it an identity?

Three sets. Type both sides; record whether every row agrees.

Left side	Right side	Equal?	A region where they differ (if any)
A – (B ∪ C)	(A – B) ∩ (A – C)	yes	none
A ∪ (B ∩ C)	(A ∪ B) ∩ (A ∪ C)	yes	none
A – (B – C)	(A – B) – C	no	in A and C, not B (element 2)
(A ∩ B)′	A′ ∪ B′	yes	none

4 Explain what did you discover?

1. The complement X′ contains everything in **U** that is not in X.
2. De Morgan: (A ∪ B)′ = **A′ ∩ B′** and (A ∩ B)′ = **A′ ∪ B′**.
3. With n sets there are **2^n** regions, so a membership table has 2^n rows.
4. Two expressions are equal sets for all A, B, C exactly when they contain the same **regions**.

5. Show with an element that $A - (B - C) \neq (A - B) - C$ for these sets, and explain why one element is enough.
   2 is in A and C but not B: $2 \in A - (B - C)$ since $2 \notin B - C$, but $2 \notin (A - B) - C$ since $2 \in C$. One counterexample disproves a claimed identity.

5 Apply a new situation, no tool

1. $U = \{1, \ldots, 10\}$, $P = \{2, 3, 5, 7\}$, $Q = \{1, 3, 5, 7, 9\}$. Find $P \cap Q'$ and $(P \cup Q)'$.
   $P \cap Q' = \{2\}$; $(P \cup Q)' = \{4, 6, 8, 10\}$.
2. $n(A) = 40$, $n(B) = 25$, $n(A \cap B) = 10$. Find $n(A \cup B)$ and $n(A \oplus B)$.
   $n(A \cup B) = 55$; $n(A \oplus B) = 45$.
3. Simplify $(A \cap B) \cup (A \cap B')$.
   A.
4. Draw a Venn diagram shading $(A \cup C) \cap B'$.
   The parts of A and C outside B: "in A only", "in C only" and "in A and C, not B".

Teacher notes

Intended discovery: Set expressions are determined by the regions they contain; equal region columns (membership table) prove an identity for all sets, and one differing region (with an element) disproves it; complements depend on U.

Likely misconceptions:

- Distributing complement over union without switching to intersection.
- Forgetting the region outside all sets in complements.
- Treating − as symmetric ($A - B = B - A$).

Answers that can vary: Equivalent expressions vary (for example $A \oplus B$ written two ways). In Record row 3, any element in "A and C, not B" (only 2 in this U) is a correct counterexample.